INCH-POUND

A-A-60012
13 December 2019

COMMERCIAL ITEM DESCRIPTION

LAMPS, INCANDESCENT, BULBS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) provides the acquisition requirements for the purchase of incandescent lamps for primarily in military aircraft applications where equipment and aircraft must operate under severe performance, storage and environmental conditions. Typical lamp usages include applications in instruments and panel illumination, control and signal indicators, anti-collision lights, wing position lights, taxiing and landing lights.

2. CLASSIFICATION. The lamps shall be classified by the following styles, rated currents, and spherical candlepowers (SCP) as shown in table I for the Part or identification number (PIN):

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

AMSC N/A FSC 6240

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TABLE I. PIN designation data for lamps.

CID PIN	Figures	Size	Rated current (±10%)	Voltage (Nom)	Watts (max)	Spherical candlepower (SCP)	Base Type	Color	Lamp life	Filament
-001	1, 4	T-1L	0.060	5.0	0.33	0.03	a.	Clear	30,000	C-2R
-002	1, 4	T-1L	0.060	5.0	0.33	0.05		Clear	20,000	C-2R
-003	1, 4	T-1L	0.060	5.0	0.33	0.05±25%		Red	20,000	C-2R
-004	1, 4	T-1L	0.115	5.0	0.63	0.15		Clear	20,000	C-2R
-005	1, 4	T-1L	0.060	5.0	0.33	0.15		Clear	2,000	C-2R
-006	2, 4	T-1S	0.060	5.0	0.33	0.03		Clear	50,000	C-2R
-007	2, 4	T-1S	0.060	5.0	0.33	0.05		Clear	20,000	C-2R
-008	2, 4	T-1S	0.075	5.0	0.41	0.09		Clear	20,000	C-2R
-009	2, 4	T-1S	0.115	5.0	0.63	0.15		Clear	20,000	C-2R
-010	2, 4	T-1S	0.115	5.0	0.63	0.15±25%		Red	20,000	C-2R
-011	2, 4	T-1S	0.060	5.0	0.33	0.15		Clear	2,000	C-2R
-012	2, 4	T-1S	0.020	28.0	0.62	0.15±25%		Clear	1,000	CC-2F
-013	3, 4	T-3/4	0.06	5.0	0.33	0.03		Clear	40,000	C-28
-014	3, 4	T-3/4	0.06	5.0	0.33	0.05		Clear	20,000	C-28
-015	3, 4	T-3/4	0.115	5.0	0.63	0.15		Clear	20,000	C-28
-016	3, 4	T-3/4	0.06	5.0	0.33	0.15		Clear	2,000	C-28

2.1 Size. The lamps shall be one of the sizes listed below.

T-3/4. As shown in figure 3.
T-1S. As shown in figure 2.
T-1L. As shown in figure 1.

2.2 Rated current. The lamp shall have one of the following designations for rated current in amperes: .020, .060, .075 or .115

2.3 Voltage. The lamps shall be capable of operating with a voltage of 5 or 28.

2.4 Spherical candlepower (SCP). The lamp's SCP shall be one of the following: .03, .05, .09, .115 or .15

2.5 Base type. The lamps shall have a base type of the following:

a. Subminiature Unbased Wire Terminals

2.6 Lamp finish color. The lamps shall have a finish color of Clear or Red.

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2.7 Lamp life. The lamps shall have an averaged lab life of the following: 1,000; 2,000; 20,000; 30,000; 40,000 or 50,000 hours

2.8 Filament style. The lamps shall have a filament style of the following (see 7.5):

C-2R	- single helical coil
C-28	- single helical coil
CC-2F	- coiled coil or double helical coil

3. SALIENT CHARACTERISTICS

3.1 Design and construction requirements. Lamps shall be of the design, color and physical dimensions expressed in the applicable PIN (see 7.3 (a)).

3.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in section 4.2 “Test conditions” of MIL-STD-202.

3.2.1 Metals. Metals shall be corrosion resistant or treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal aircraft service. Lamp base and contact materials shall be resistant to electrolytic corrosion when installed in metal sockets and holders.

3.2.2 Lamps. Lamps shall be clear or colored incandescent lamps for use at ambient temperatures up to 85°C (185°F).

3.2.2.1 Lamp operation. Unless otherwise specified, lamps shall be operated at the rated voltage. Voltage tolerance shall be within ± 0.5 percent of the rated voltage.

3.2.2.2 Coatings. The exterior coating on lamps (colored) shall be such that it will withstand the rigors of normal handling, service use and storage.

3.2.2.3 Colored lamps. The lamps shall be uniformly colored with no perforations or openings through which non-colored light can be emitted. All light emitted from colored lamps shall conform to SAE-AS25050 for the color specified. Color may be obtained by the use of either colored glass or a coating on the glass.

3.2.3 Filament. The structure and processing of the filament material shall be of a quality to meet the specification performance and environmental requirements. The arrangement of the filament and filament support shall be as designated by the applicable filament style in the PIN (see 7.3 (a)). When not specified, the filament arrangement and support shall be at the manufacturer's option, to meet the performance and environmental requirements specified herein.

3.2.4 Lamp bases. Where applicable, lamp base configuration and dimensions shall conform to the base type designation of ANSI-C81.61 as specified in the PIN (see 7.3 (a)).

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3.2.5 Examination of product. Lamps shall be visually and dimensionally inspected for conformance to the applicable PIN (see 7.3 (a)) and visually examined for conformance to workmanship and identification of product.

3.2.6 Workmanship. Each lamp, including all parts and accessories, shall be fabricated and finished, free of blemishes and defects which will adversely affect its life, form, fit or function. Soldering, welding, brazing, cementing and wiring shall be thorough and alignment of parts shall be accurate. Each lamp shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, flux, and other foreign material shall be removed (see 3.2.5 and 7.5).

3.2.7 Ratings. When operating at rated voltage, lamps shall conform to the current, voltage, and light output ratings specified in the applicable PIN (see 7.3 (a)).

3.2.7.1 Burn-in/seasoning. Prior to the start of testing, the lamps shall be seasoned by being operated at the rated voltage for a period of time equal to one percent of the rated life specified in the applicable PIN (see 7.3 (a)) or a maximum of 10 hours.

3.2.7.2 Light output. Unless otherwise specified, light output criteria shall be as specified in the PIN. The acceptable tolerance shall be ±15% SCP. Colored lamps shall be operated at rated voltage and the light emitted shall be measured for conformance to SAE-AS25050 for the specified color.

3.2.7.2.1 Initial candlepower. Lamps furnished under this specification shall have an initial candlepower output within the range specified in the applicable PIN (see 7.3 (a)) when tested as specified (see 3.2.7).

3.2.7.2.2 Candlepower maintenance. The average candlepower output of lamps shall not fall below 80 percent of the initial candlepower output at 70 percent of average rated lab life as specified in the applicable PIN (see 7.3 (a)). The exterior coating of lamps (if so furnished) shall not be damaged or discolor, crack, fade, blister or peel (see 7.5).

3.2.8 Life. Life requirements for lamps furnished under this specification shall have at a minimum the average rated laboratory life (see 7.6), as specified in the PIN (see 7.3 (a)).

3.2.9 Coating adhesion. When tested for adherence to the glass bulb, the exterior coatings (colored), shall be such that they shall not be damaged, blistered, cracked or peeled from the globe when tested as specified (see 3.2.11.1 and 7.5). The lamps shall be capable of being tested to determine the coating adhesion by tape stripping using the following procedure. Pressure sensitive adhesive tape with a width of 0.75 inches and a minimum length of 2 inches conforming to ASTM D6123/D6123M, Type 1, shall be pressed firmly onto a flat or cylindrical coated surface of the sample lamp rubbing out all air bubbles under the tape. Ten seconds shall be allowed for the test area to return to room temperature. Then grasping a free end of the tape and in a rapid motion the tape shall be pulled back upon itself at 180 degrees (in such a manner that the tape is folded back to back during the procedure). The lamps shall be capable of passing this test both before and after the lamps are operated for 10 percent of the rated life.

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3.2.10 Solderabilitv. Unbased lamps having tinned lead terminals shall have the solder dipped surface of the terminal be at least 95 percent covered with a new, smooth solder coating when tested in accordance with MIL-STD-202-208. The remaining 5 percent of the terminal surface may contain small pinholes, voids or rough spots not concentrated in any one area. Bare base metal and areas where the new solder dip failed to cover the original coating are indications of poor solderability.

3.2.11 Environmental. Lamps shall perform within the specified rating limits when subjected to the following environmental requirements for the following specified times, temperatures and environmental conditions, unless otherwise specified.

3.2.11.1 High temperature. When operating at rate voltage, the lamps shall perform to a minimum of 70 percent of the average rated life specified in the applicable PIN (see 7.3 (a), up to 1,000 hours maximum test time) while exposed to a minimum temperature of 85°C (185°F). Lamps shall show no evidence of burn-out or damage to any lamp component as a result of exposure to temperature extremes. Exterior coated lamps (colored) shall show no evidence of damage, discoloration, cracking, fading, blistering or peeling of the coating (see 7.5).

3.2.11.1.1 Precautions. The following precautions shall be observed when performing the high temperature tests:

a. The test chamber shall be capable of maintaining the specified temperature within ± 5°C (± 9°F). The test chamber shall be operated for one hour prior to installation of lamps to ensure that the temperature has stabilized.

b. Temperature measurements shall be made with thermocouples having leads at least 20 inches long within the oven. This will minimize the conduction of heat away from the junction.

c. Baffles shall be placed within the oven so as to shield the lamps and thermocouple junctions from direct radiation from the heating elements and surfaces of the oven.

d. The lamps shall be mounted so that there is a clear space between the lamps equal to at least the width of one lamp. The lamps shall be mounted so that the flux from any lamp does not directly strike any other lamp.

3.2.11.2 Thermal shock. When operating at rated voltage, the lamps shall show no evidence of burn-out or damage to any lamp component when subjected to severe changes in thermal conditions as specified in MIL-STD-202-107; the exceptions indicted in table II shall replace table I of MIL-STD-202-107. Continue steps 1 through 4 of table II for 25 cycles. Exterior coated lamps (colored) shall show no evidence of damage, discoloration, cracking, fading, blistering or peeling of the coating (see 7.5).

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TABLE II. Thermal shock test conditions.

Step	Lamps		
	Temperature		Time per cycle
	°C	°F	
1	-55+1	-67+2	Table II
2	+25+2	+77+4	5 minutes max
3	+85+2	+185+4	Table II
4	+25+2	+77+4	5 minutes max

1/ High temperature as specified in 3.2.2 .

3.2.11.3 Random vibration. When operating at rated voltage, the lamps shall show no evidence of damage to any lamp component, or loose parts (see 7.5) when subjected to random vibrations in a frequency range from 5 to 2,000 Hz with power spectral densities as specified in the applicable PIN (see 7.3 (a)) for each lamp type. The vibration apparatus control and analysis of vibration shall be in accordance with MIL-STD-202-214. For vibration tests, the lamps shall be potted in a clear epoxy block which shall be rigidly mounted to the vibration table.

3.2.11.3.1 Test setup applicable to all lamps. The lamps shall be capable of being vibrated for a total of at least 30 minutes; 7.5 minutes with the lamps non-operating and 22.5 minutes with the lamps operating at rated voltage. Non-operating vibration shall be in the "Z" axis. During the operating mode, the lamps shall be vibrated for 7.5 minutes in each of the 3 orthagonal axes. The vibration with lamps operating shall start immediately after lamps are lit without any additional burn-in. Either the fixture or the lamps may be rotated to achieve the positioning of lamps in each of the axes. After this test, the lamps shall conform to 3.2.11.7 .

3.2.11.4 Shock. When operating at rated voltage, the lamps shall show no evidence of burn-out, damage to any lamp component, or loose parts (see 7.5) when subjected to a shock test using Test Condition J, MIL-STD-202-213 with the exceptions indicated herein. This test shall be performed with the lamps operating at the rated voltage. A peak "g" load value as specified in the applicable PIN (see 7.3 (a)) shall be substituted for that specified in Test Condition J, Table 1.

3.2.11.5 Humidity. The lamps, non-operating, shall be capable of being subjected to a humidity test using Condition A (240 hours), MIL-STD-202-103. After exposure to the hot and humid atmosphere, the lamps shall be capable of being visually inspected for globe to base integrity and any evidence of atmospheric leakage into the lamp. The lamps shall then be capable of being operated at rated voltage to determine conformance to the ratings (current, power and light output) specified in the applicable PIN (see 7.3 (a)).

3.2.11.6 Salt spray. The lamps, non-operating, shall be capable of being subjected to a salt spray test using Test Condition B (48 hours), MIL-STD-202-101 without visual signs of corrosion following exposure. The lamps shall then be capable of being operated at rated voltage to determine conformance to the ratings (current, power and light output) specified in the applicable PIN (see 7.3 (a)).

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3.3 Identification of product. Unless otherwise specified, all lamps shall be marked for identification in accordance with the marking specified in 6.1 . Markings shall be clear, legible and durable (see 3.2.5).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance, and to perform, at its own expense, those additional inspections, tests and other evaluations the contracting agency deems necessary.

5.2 First article inspection. When specified (see note 7.3 (c)), a first article sample shall be inspected to verify conformance to the salient characteristics in this CID. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply items that are fully representative of those inspected as a first article sample.

5.3 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order (see 7.3 (f)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

AA60012 - XXX

See TABLE I.

CID number

7.2 Sources of documents.

7.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks listed below form a part of this specification to the extent specified herein.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202
MIL-STD-202-101
MIL-STD-202-103
MIL-STD-202-107
MIL-STD-202-208
MIL-STD-202-213
MIL-STD-202-214

(Copies of these documents are available at https://quicksearch.dla.mil/.)

7.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D6123/D6123M

(Copies of these documents are available at https://www.astm.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C81.61

(Copies of these documents are available at https://www.nema.org.)

SAE INTERNATIONAL

SAE-AS25050

(Copies of these documents are available at https://www.sae.org.)

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7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN (see table I).
b. Color of lamp required, if not otherwise specified in the PIN.
c. If first article is required and arrangements for first article inspection (5.2).
d. Sampling plans and acceptance levels for inspections.
e. Sampling and inspection requirements for packaging (6).
f. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

Aircraft lighting
Illumination
Lighting
Lights

7.5 Definitions. For purposes of defining various types of defects or terminology which is referred to in this specification or in the applicable PIN, the following definitions apply:

Blemish - A visual nonfunctional imperfection of the symmetry of the lamp globe or base.

Blister - A visible separation of a coating from the glass bulb which forms an upraised, unbroken bubble.

Crack - A narrow lengthwise opening, split or fissure produced by a partial separation of the coating from the bulb surface.

Damage - Any lamp component defect which prohibits the lamp from conforming to its basic functional and light distribution requirements.

Defect - A fault in workmanship or manufacture which affects the proper functioning of the item.

Discolor - A distinct change in hue of color or coating which adversely affects the lamp's functional color or reflective qualities and reduces its illumination characteristics to an out of specification condition.

Fade - A loss of brightness or vividness of color such as to adversely affect the lamp's functional color or reflective qualities and reduces its illumination characteristics to an out of specification condition.

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Filament - C = single helical coil
CC = coiled coil or double helical coil
F = flat coil
S = straight uncoiled wire

Light center for C-2 shaped filaments - Light center is a point lying in a plane through the axis of the filament coils midway between the "top" and "bottom" of the lighted filament and midway between the outer edges of the filament coils at the horizontal level of the light center.
NOTE: Filament top is the upper extremity of the highest lighted turn (away from base) in either coil segment. Filament bottom is the underside of the lowest lighted turn (toward base) in either coil segment. The lowest lighted turn is considered for this purpose as being the first turn above the first inside angle between turns of the projected filament image.

7.6 Rated life. The average laboratory lamp life is determined by operation of the lamps at rated voltage. The average represents the number of hours that it would take for 50 percent of the test samples to fail. Lamps are considered as having failed when they burn out or light output decreases by more than 20 percent of initial reading. The specified average rated life (see 3.2.8) is based upon operating the lamps at the specified design voltage. Where performance of life testing at rated voltage is considered to be impractical, the use of accelerated life testing based on the 12th power rule is permissible.

$$L_A = L_R(V_R/V_O)^{12}$$

Where L_A = Accelerated life
L_R = Rated life at design voltage
V_R = Rated voltage
V_O = Operating voltage

Upon application to, and approval of the acquiring activity, the manufacturer may provide test results from their in-process quality conformance testing to substantiate the life requirements.

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7.7 Figures.

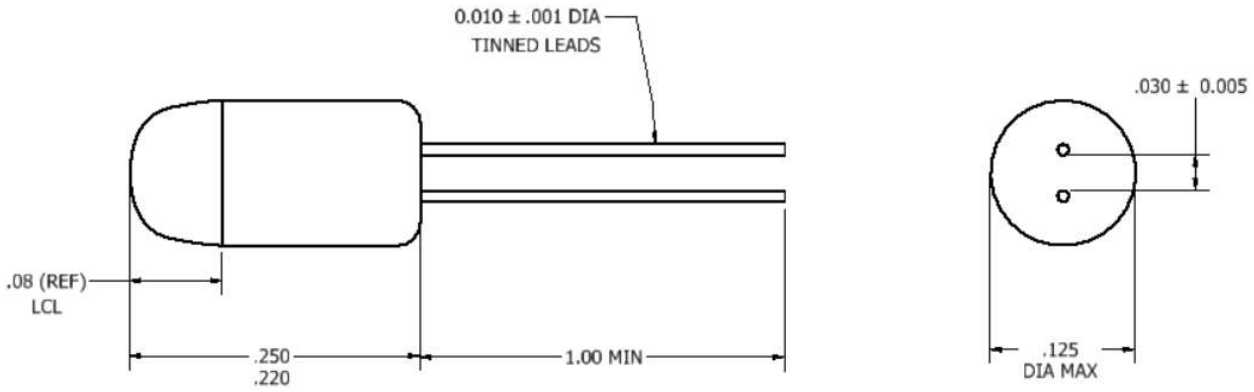


FIGURE 1. T-1 long configuration and dimensions.

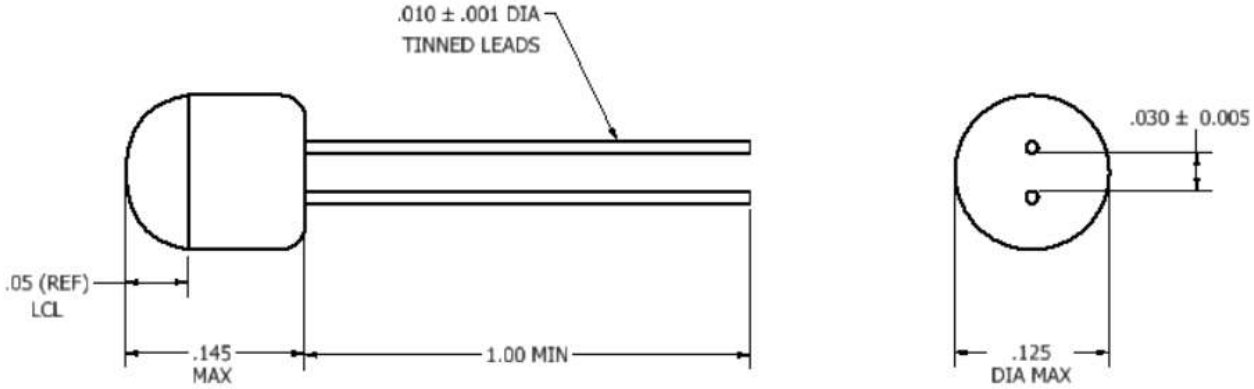


FIGURE 2. T-1 short configuration and dimensions.

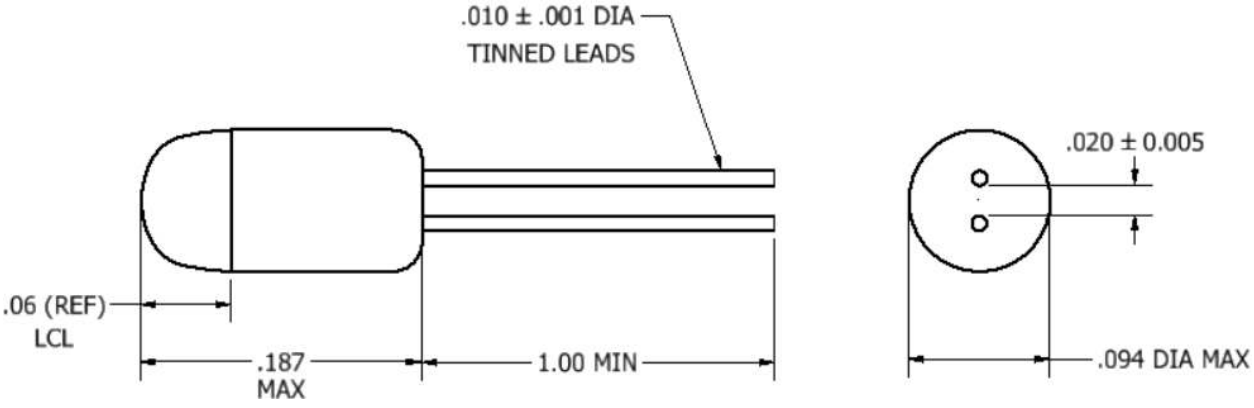
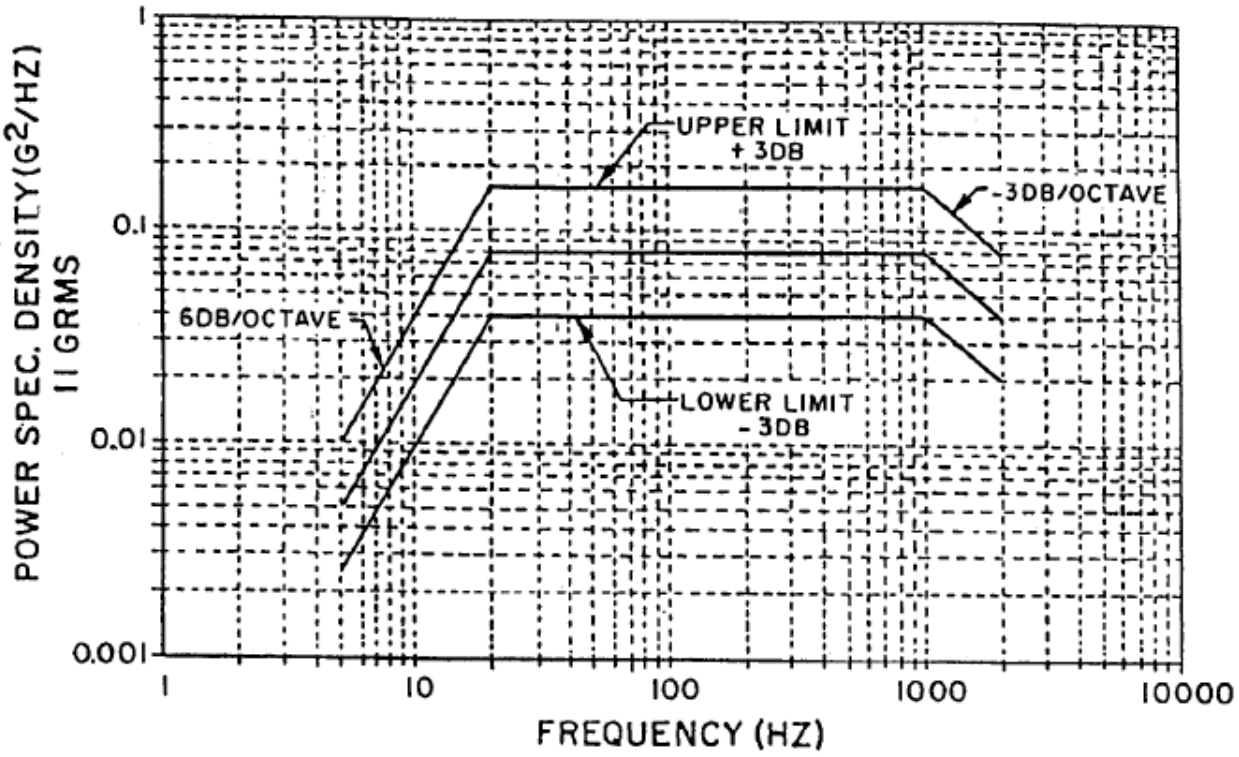


FIGURE 3. T-3/4 configuration and dimensions.

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Freq Hz	Vibration Level PSD (G^2/Hz)	Upper Limit (+3 DB) PSD (G^2/Hz)	Lower Limit (-3 DB) PSD (G^2/Hz)
5	0.005	0.01	0.0025
10	0.02	0.04	0.01
20	0.08	0.16	0.04
1000	0.08	0.16	0.04
2000	0.04	0.08	0.02

FIGURE 4. Random vibration curve for T- ¾ and T-1 lamps.

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TABLE III. CID reference.

CID PIN	MIL-DTL-6363 similarity
AA60012-001	M6363/5-1 AS15
AA60012-002	M6363/5-2 AS15
AA60012-003	M6363/5-2R
AA60012-004	M6363/5-3 AS15
AA60012-005	M6363/5-4 AS15
AA60012-006	M6363/5-5 AS15
AA60012-007	M6363/5-6 AS15
AA60012-008	M6363/5-7 AS15
AA60012-009	M6363/5-8 AS15
AA60012-010	M6363/5-8R
AA60012-011	M6363/5-9 AS15
AA60012-012	M6363/5-10
AA60012-013	M6363/4-1
AA60012-014	M6363/4-2
AA60012-015	M6363/4-3
AA60012-016	M6363/4-4

MILITARY INTERESTS:

Custodians:
Army - AV
Navy - SH
Air Force - 11
DLA - GS

Review Activities:
Air Force - 71
Navy - AS, MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA – GS2
(Project 6240-2020-002)

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